Fair Class-Incremental Learning using Sample Weighting

Jaeyoung Park*
Korea Advanced Institute of Science and Technology
Daejeon, Republic of Korea
jypark@kaist.ac.kr

Minsu Kim*
Korea Advanced Institute of Science and Technology
Daejeon, Republic of Korea
ms716@kaist.ac.kr

Steven Euijong Whang†
Korea Advanced Institute of Science and Technology
Daejeon, Republic of Korea
swhang@kaist.ac.kr

Abstract

Model fairness is becoming important in class-incremental learning for Trustworthy AI. While accuracy has been a central focus in class-incremental learning, fairness has been relatively understudied. However, naïvely using all the samples of the current task for training results in *unfair catastrophic forgetting* for certain sensitive groups including classes. We theoretically analyze that forgetting occurs if the average gradient vector of the current task data is in an "opposite direction" compared to the average gradient vector of a sensitive group, which means their inner products are negative. We then propose a *fair class-incremental learning* framework that adjusts the training weights of current task samples to change the direction of the average gradient vector and thus reduce the forgetting of underperforming groups and achieve fairness. For various group fairness measures, we formulate optimization problems to minimize the overall losses of sensitive groups while minimizing the disparities among them. We also show the problems can be solved with linear programming and propose an efficient Fairness-aware Sample Weighting (FSW) algorithm. Experiments show that FSW achieves better accuracy-fairness tradeoff results than state-of-the-art approaches on real datasets. The source code is released at https://github.com/jyparkkr/FSW.

CCS Concepts

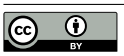
• **Computing methodologies** → **Machine learning**.

Keywords

trustworthy ai, model fairness, continual learning, class-incremental learning

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*Equal contribution.
†Corresponding author.



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1 Introduction

Trustworthy AI is becoming critical in various continual learning applications including autonomous vehicles, personalized recommendations, healthcare monitoring, and more [25, 33]. Improving both model fairness and accuracy is crucial, as unfair predictions can significantly compromise trust and safety, particularly in human-centered automated systems, especially as observed frequently in the context of continual learning. In this paper, we focus on class-incremental learning, where the objective is to incrementally learn new classes as they appear.

The main challenge of class-incremental learning is to learn new classes of data, while not forgetting previously-learned classes [6, 31]. If we simply fine-tune the model on the new classes, the model will gradually forget about the previously-learned classes. This phenomenon called catastrophic forgetting [30, 38, 44] may easily occur in real-world scenarios where the model needs to continuously learn new classes. Therefore, we need to have a balance between learning new information and retaining previously-learned knowledge, which is called the stability-plasticity dilemma [1, 28, 40].

In this paper, we solve the problem of *fair class-incremental learning* where the goal is to satisfy various notions of fairness among sensitive groups including classes in addition to classifying accurately. In continual learning, unfair forgetting may occur if the current task data has similar characteristics to previous data, but belongs to different sensitive groups including classes, which negatively affects the performance on the previous data during training. Despite the importance of the problem, the existing research [13, 52] is still nascent and has limitations in terms of technique or scope (see Sec. 2). In comparison, we support fairness more generally in class-incremental learning by satisfying various notions of group fairness for sensitive groups including classes.

We demonstrate how unfair forgetting can occur on a synthetic dataset with two attributes (x_1, x_2), and one true label y as shown in Fig. 1a. We sample data for each class from three normal distributions: $(x_1, x_2)|y = 0 \sim \mathcal{N}([-2; -2], [1; 1])$, $(x_1, x_2)|y = 1 \sim \mathcal{N}([2; 4], [1; 1])$, and $(x_1, x_2)|y = 2 \sim \mathcal{N}([4; 2], [1; 1])$. To simulate class-incremental learning, we introduce data for Class 0 (blue) and Class 1 (orange) in Task 1, followed by Class 2 (green) data in Task 2, where Class 2's data is similar to Class 1's data. We observe that this setting frequently occurs in real datasets, where different classes of data exhibit similar features or characteristics, as shown in Sec. B.1 of our full paper [45]. We assume a data replay setting in which only a limited amount of data from Classes 0 and 1 is stored and jointly used during training on Class 2. After completing training on Task 1, we observe how the model's accuracy on the three classes changes during Task 2 training in Fig. 1b. As the accuracy on Class 2 improves, there is a catastrophic forgetting of Class 1 only, which leads to unfairness.

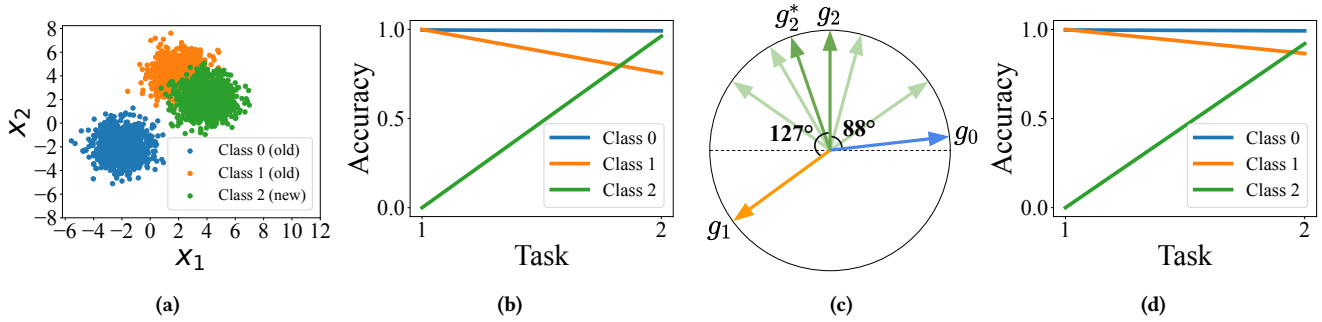


Figure 1: (a) A synthetic dataset for class-incremental learning. (b) Training on Class 2 results in unfair forgetting on Class 1 only. (c) The average gradient vector of Class 2, g_2, is more than $90°$ apart from Class 1's g_1, which means the model is being trained in an opposite direction. Our method adjusts g_2 to g_2^* through sample weighting to be closer to g_1, but not too far from the original g_2. (d) As a result, the unfair forgetting is mitigated while minimally sacrificing accuracy for Class 2.

Beyond the label, unfairness can also arise when forgetting occurs among groups defined by both label y and a sensitive attribute z such as age, gender, or race. For instance, we can consider Class 0 as $(y = 0, z = 0)$, Class 1 as $(y = 0, z = 1)$, and Class 2 as $(y = 1, z = 1)$. Introducing Class 2 in Task 2 can lead to an accuracy drop on Class 1 $(z = 1)$, while Class 0 $(z = 0)$ remains unaffected. Likewise, unfairness can arise when groups are defined by both class labels and sensitive attributes.

To analytically understand the unfair forgetting, we project the average gradient vector for each class data on a 2-dimensional space in Fig. 1c. Here g_0, g_1, and g_2 represent the average gradient vectors of the samples of Classes 0, 1, and 2, respectively. We observe that g_2 is $127°$ apart from g_1, but $88°$ from g_0, which means that the inner products $\langle g_2, g_1 \rangle$ and $\langle g_2, g_0 \rangle$ are negative and close to 0, respectively. In Sec. 3.1, we theoretically show that a negative inner product between average gradient vectors of current and previous data results in higher loss for the previous data as the model is being updated in an opposite direction and identify a sufficient condition for unfair forgetting. As a result, Class 1's accuracy decreases, while Class 0's accuracy remains stable.

Our solution to mitigate unfair forgetting is to adjust the average gradient vector of the current task data by weighting its samples. The light-green vectors in Fig. 1c are the gradient vectors of individual samples from Class 2, and by weighting them we can adjust g_2 to g_2^* to make the inner product with g_1 less negative. At the same time, we do not want g_2^* to be too different from g_2 and lose accuracy. In Sec. 3.2, we formalize this idea using the weighted average gradient vector of the current task data. We then optimize the sample weights such that unfair forgetting and accuracy reduction over sensitive groups including classes are both minimized. We show this optimization can be solved with linear programming and propose our efficient Fairness-aware Sample Weighting (FSW) algorithm. Fig. 1d shows how using FSW mitigates the unfair forgetting between Classes 0 and 1 without harming Class 2's accuracy much. Our framework supports the group fairness measures equal error rate [53], equalized odds [21], and demographic parity [20] and can be potentially extended to other measures.

In our experiments, we show that FSW achieves better fairness and competitive accuracy compared to state-of-the-art baselines on various image, text, and tabular datasets. The benefits come from assigning different training weights to the current task samples with accuracy and fairness in mind.

Summary of Contributions: (1) We theoretically analyze how unfair catastrophic forgetting can occur in class-incremental learning; (2) We formulate optimization problems for mitigating the unfairness for various group fairness measures and propose an efficient fairness-aware sample weighting algorithm, FSW; (3) We demonstrate how FSW outperforms state-of-the-art baselines in terms of fairness with comparable accuracy on various datasets.

2 Related Work

Class-incremental learning is a challenging type of continual learning where a model continuously learns new tasks, each composed of new disjoint classes, and the goal is to minimize catastrophic forgetting [36, 37]. Data replay techniques [12, 34, 47] store a small portion of previous data in a buffer to utilize for training and are widely used with other techniques (see more details in Sec. C of our full paper [45]). There are also more advanced gradient-based sample selection techniques like GSS [3] and OCS [58] that manage buffer data to have samples with diverse and representative gradient vectors. All these works do not consider fairness and simply assume that the entire incoming data is used for model training, which may result in unfair forgetting as we show in our experiments.

Model fairness research mitigates bias by ensuring that a model's performance is equitable across different sensitive groups, thereby preventing discrimination based on race, gender, age, or other sensitive attributes [39] (see more details in Sec. C of our full paper [45]). However, most of these techniques assume that the training data is given all at once, which may not be realistic. There are techniques for fairness-aware active learning [4, 43, 51], in which the training data evolves with the acquisition of samples. However, these techniques store all labeled data and use them for training, which is impractical in continual learning settings.

A recent study addresses model fairness in class-incremental learning, where models may suffer from imbalanced forgetting of previously learned sensitive groups including classes, leading to unfairness across different groups [13, 22, 52, 55] (see more details

in Sec. C of our full paper [45]). FaIRL [13] supports group fairness metrics like demographic parity for continual learning, but proposes a representation learning method that does not directly optimize the given fairness measure and thus has limitations in improving fairness as we show in experiments. FairCL [52] also addresses fairness in a continual learning setup, but only focuses on resolving the imbalanced class distribution based on the number of pixels of each class in an image, and is specific to semantic segmentation tasks. In comparison, we can mitigate bias from both data imbalance and intrinsic or acquired characteristics within the data by satisfying multiple notions of group fairness for sensitive groups including classes.

3 Framework

In this section, we first theoretically analyze unfair forgetting using gradient vectors of sensitive groups and the current task data. Next, we propose a sample weighting algorithm to mitigate unfairness by adjusting the average gradient vector of the current task data.

Notations. In class-incremental learning, a model incrementally learns new current task data along with previous buffer data using data replay. Suppose we train a model to incrementally learn L tasks $\{T_1, T_2, \ldots, T_L\}$ over time, and there are N classes in each task with no overlapping classes between different tasks. After learning the lth task T_l, we would like the model to remember all $(l-1)\cdot N$ previous task classes and an additional N current task classes. We assume the buffer has a fixed size of M samples. For L tasks, we allocate $m = M/L$ samples of buffer data per task. If each task consists of N classes, then we allocate $m/N = M/(L\cdot N)$ samples of buffer data per class [10, 11, 41]. Each task $T_l = \{d_i = (X_i, y_i)\}$ is composed of feature-label pairs. We also use $\mathcal{M}_l = \{d_j = (X_j, y_j)\}_{j=1}^{m}$ to represent the buffer data for each previous l^{th} task T_l. We assume the buffer data per task is small, i.e., $m \ll |T_l|$ [12].

When defining fairness for class-incremental learning, we utilize sensitive groups including classes. According to the fairness literature, sensitive groups are divided by sensitive attributes like gender and race (e.g., Male and Female). Similarly, the classes can also be viewed as sensitive groups, where the class label serves as the sensitive attribute. To support either case in a class-incremental setting, we use the following unifying notations: (1) if the sensitive groups are classes, then they form the set $G_y = \{(X, \mathrm{y}) \in \mathcal{D} : \mathrm{y} = y, y \in \mathbb{Y}\}$ where $\mathcal{D}$ is a dataset, y is a class attribute, and $\mathbb{Y}$ is the set of y; (2) if we are using sensitive attributes beyond classes, we can further divide the classes into the set $G_{y,z} = \{(X, \mathrm{y}, \mathrm{z}) \in \mathcal{D} : \mathrm{y} = y, \mathrm{z} = z, y \in \mathbb{Y}, z \in \mathbb{Z}\}$ where z is a sensitive attribute and $\mathbb{Z}$ is the set of z. See Sec. A.1 of our full paper [45] for a summary of notations used.

3.1 Unfair Forgetting

We take inspiration from GEM [34], which theoretically analyzes catastrophic forgetting by utilizing the angle between gradient vectors of data. If the inner products of gradient vectors for previous tasks and the current task are negative (i.e., $90^\circ < \text{angle} \leq 180^\circ$), the loss of previous tasks increases after learning the current task. Catastrophic forgetting thus occurs when the gradient vectors of different tasks point in opposite directions. Intuitively, the opposite gradient vectors update the model parameters in conflicting directions, leading to forgetting while learning.

Using the notion of catastrophic forgetting, we propose theoretical results for unfair forgetting:

Lemma 3.1. *Denote G as a sensitive group containing features X and true labels y. Also, denote f_θ^{l-1} as a previous model and f_θ as the updated model after training on the current task T_l. Let ℓ be any differentiable loss function (e.g., cross-entropy loss), and η be a learning rate. Then, the loss of the sensitive group of data after training with a current task sample $d_i \in T_l$ is approximated as:*

$$\tilde{\ell}(f_\theta, G) = \ell(f_\theta^{l-1}, G) - \eta \nabla_\theta \ell(f_\theta^{l-1}, G)^\top \nabla_\theta \ell(f_\theta^{l-1}, d_i), \quad (1)$$

where $\tilde{\ell}(f_\theta, G)$ is the approximated average loss between model predictions $f_\theta(X)$ and true labels y, whereas $\ell(f_\theta^{l-1}, G)$ is the exact average loss, $\nabla_\theta \ell(f_\theta^{l-1}, G)$ is the average gradient vector for the samples in the group G, and $\nabla_\theta \ell(f_\theta^{l-1}, d_i)$ is the gradient vector for a sample d_i, each with respect to the previous model f_θ^{l-1}.

The proof is in Sec. A.2 of our full paper [45]. We employ first-order Taylor series approximation for the proof, which is widely used in the continual learning literature, by assuming that the loss function is locally linear in small optimization steps and considering the first-order term as the cause of catastrophic forgetting [3, 32, 34]. We empirically find that the approximation error is large when a new task begins because new samples with unseen classes are introduced. However, the error gradually becomes quite small as the number of epochs increases while training a model for the task, as in Sec. B.2 of our full paper [45]. Additionally, while higher-order terms can be included in principle, doing so makes the optimization non-convex and NP-hard. A detailed proof is provided in Sec. A.3 of our full paper [45].

To define fairness in class-incremental learning with the approximated loss, we adopt the definition of approximate fairness that considers a model to be fair if it has approximately the same loss on the positive class, independent of the group membership [18]. In this paper, we compute fairness measures based on the disparity between approximated cross-entropy losses, which are derived from Lemma 3.1 using gradients. The following proposition shows how using the cross-entropy loss disparity can effectively approximate common group fairness metrics such as equalized odds and demographic parity. We choose cross-entropy loss specifically because it is widely validated as a strong proxy for fairness disparity metrics, though our method is robust to alternative loss choices (see Sec. A.4 and B.14 of our full paper [45] for more justification of the loss function and an alternative, respectively).

Proposition 3.2. *(From [48–50]) Using the cross-entropy loss disparity to measure fairness is empirically verified to provide reasonable proxies for common group fairness metrics.*

Based on Lemma 3.1 and Proposition 3.2, Theorem 3.3 provides a sufficient condition for unfair forgetting. Intuitively, if a sample's gradient opposes only the underperforming group's average gradient, the training amplifies unfairness between the two groups.

Theorem 3.3. *Let ℓ be the cross-entropy loss and G_1 and G_2 the overperforming and underperforming sensitive groups of data, respectively. Also let d_i be a training sample that satisfy the following conditions: $\ell(f_\theta^{l-1}, G_1) < \ell(f_\theta^{l-1}, G_2)$ while $\nabla_\theta \ell(f_\theta^{l-1}, G_1)^\top \nabla_\theta \ell(f_\theta^{l-1}, d_i) > 0$ and $\nabla_\theta \ell(f_\theta^{l-1}, G_2)^\top \nabla_\theta \ell(f_\theta^{l-1}, d_i) < 0$. Then $|\tilde{\ell}(f_\theta, G_1) - \tilde{\ell}(f_\theta, G_2)| > |\ell(f_\theta^{l-1}, G_1) - \ell(f_\theta^{l-1}, G_2)|$.*

The proof is in Sec. A.2 of our full paper [45]. The result shows that the loss disparity between the two groups could become larger after training on the current task sample, which leads to worse fairness. This theorem can be extended to when we have a set of current task samples $T_l = \{d_i = (X_i, y_i)\}$ where we can replace $\nabla_\theta \ell(f_\theta^{l-1}, d_i)$ with $\frac{1}{|T_l|} \sum_{d_i \in T_l} \nabla_\theta \ell(f_\theta^{l-1}, d_i)$. If the average gradient vector of the current task data satisfies the derived sufficient condition, training with all of the current task samples using equal weights could thus result in unfair catastrophic forgetting.

3.2 Sample Weighting for Unfairness Mitigation

To mitigate unfairness, we propose sample weighting to suppress samples that harm fairness and promote samples that help. Given training weights $\mathbf{w}_l \in [0, 1]^{|T_l|}$ for the samples in the current task data, the approximated loss of a group G after training is:

$$\tilde{\ell}(f_\theta, G) = \ell(f_\theta^{l-1}, G) - \eta \nabla_\theta \ell(f_\theta^{l-1}, G)^\top \left(\frac{1}{|T_l|} \sum_{d_i \in T_l} \mathbf{w}_l^i \nabla_\theta \ell(f_\theta^{l-1}, d_i) \right),$$

where $\mathbf{w}_l^i$ is a training weight for the current task sample d_i. We then formulate an optimization problem to find the weights such that both loss and unfairness are minimized. Here we define $\mathbb{Y}$ as the set of all classes and $\mathbb{Y}_c$ as the set of classes in the current task. We represent accuracy as the average loss over the current task data and minimize the cost function $\mathcal{L}_{acc} = \tilde{\ell}(f_\theta, G_{\mathbb{Y}_c}) = \frac{1}{|\mathbb{Y}_c|} \sum_{y \in \mathbb{Y}_c} \tilde{\ell}(f_\theta, G_y) = \frac{1}{|\mathbb{Y}_c||\mathbb{Z}|} \sum_{y \in \mathbb{Y}_c, z \in \mathbb{Z}} \tilde{\ell}(f_\theta, G_{y,z})$. For fairness, the cost function $\mathcal{L}_{fair}$ depends on the group fairness measure as we explain below. We then minimize $\mathcal{L}_{fair} + \lambda \mathcal{L}_{acc}$ where λ is a hyperparameter that balances fairness and accuracy.

Equal Error Rate (EER). This measure [53] is defined as $\Pr(\hat{y} \neq y_1 | y = y_1) = \Pr(\hat{y} \neq y_2 | y = y_2)$ for $y_1, y_2 \in \mathbb{Y}$, where $\hat{y}$ is the predicted class, and y is the true class. We define the cost function for EER as the average absolute deviation of the class loss, consistent with the definition of group fairness metrics: $\mathcal{L}_{EER} = \frac{1}{|\mathbb{Y}|} \sum_{y \in \mathbb{Y}} |\tilde{\ell}(f_\theta, G_y) - \tilde{\ell}(f_\theta, G_{\mathbb{Y}})|$. The entire optimization problem is:

$$\begin{aligned} \min_{\mathbf{w}_l} \frac{1}{|\mathbb{Y}|} \sum_{y \in \mathbb{Y}} |\tilde{\ell}(f_\theta, G_y) - \tilde{\ell}(f_\theta, G_{\mathbb{Y}})| \; + \lambda \frac{1}{|\mathbb{Y}_c|} \sum_{y \in \mathbb{Y}_c} \tilde{\ell}(f_\theta, G_y), \\ \text{where } \; \tilde{\ell}(f_\theta, G_y) = \ell(f_\theta^{l-1}, G_y) \\ -\eta \nabla_\theta \ell(f_\theta^{l-1}, G_y)^\top \left(\frac{1}{|T_l|} \sum_{d_i \in T_l} \mathbf{w}_l^i \nabla_\theta \ell(f_\theta^{l-1}, d_i) \right). \end{aligned} \tag{2}$$

Equalized Odds (EO). This measure [21] is satisfied when sensitive groups have the same accuracy, i.e., $\Pr(\hat{y} = y | y = y, z = z_1) = \Pr(\hat{y} = y | y = y, z = z_2)$ for $y \in \mathbb{Y}$ and $z_1, z_2 \in \mathbb{Z}$. We design the cost function for EO as $\mathcal{L}_{EO} = \frac{1}{|\mathbb{Y}||\mathbb{Z}|} \sum_{y \in \mathbb{Y}, z \in \mathbb{Z}} |\tilde{\ell}(f_\theta, G_{y,z}) - \tilde{\ell}(f_\theta, G_y)|$, and the entire optimization problem is:

$$\begin{aligned} \min_{\mathbf{w}_l} \frac{1}{|\mathbb{Y}||\mathbb{Z}|} \sum_{y \in \mathbb{Y}, z \in \mathbb{Z}} |\tilde{\ell}(f_\theta, G_{y,z}) - \tilde{\ell}(f_\theta, G_y)| \\ + \lambda \frac{1}{|\mathbb{Y}_c||\mathbb{Z}|} \sum_{y \in \mathbb{Y}_c, z \in \mathbb{Z}} \tilde{\ell}(f_\theta, G_{y,z}), \\ \text{where } \; \tilde{\ell}(f_\theta, G_{y,z}) = \ell(f_\theta^{l-1}, G_{y,z}) \\ -\eta \nabla_\theta \ell(f_\theta^{l-1}, G_{y,z})^\top \left(\frac{1}{|T_l|} \sum_{d_i \in T_l} \mathbf{w}_l^i \nabla_\theta \ell(f_\theta^{l-1}, d_i) \right). \end{aligned} \tag{3}$$

Demographic Parity (DP). This measure [20] is satisfied by minimizing the difference in positive prediction rates between sensitive groups. Here, we extend the notion of demographic parity to the multi-class setting [2, 16], i.e., $\Pr(\hat{y} = y | z = z_1) = \Pr(\hat{y} = y | z = z_2)$ for $y \in \mathbb{Y}$ and $z_1, z_2 \in \mathbb{Z}$. In the binary setting of $\mathbb{Y} = \mathbb{Z} = \{0, 1\}$, a sufficient condition for demographic parity is suggested using the loss multiplied by the ratios of sensitive groups [48]. By extending to multi-class, we derive a sufficient condition as follows: $\frac{m_{y,z_1}}{m_{*,z_1}} \ell(f_\theta, G_{y,z_1}) = \frac{m_{y,z_2}}{m_{*,z_2}} \ell(f_\theta, G_{y,z_2})$ where $m_{y,z} := |\{i : y_i = y, z_i = z\}|$ and $m_{*,z} := |\{i : z_i = z\}|$. The proof is in Sec. A.5. Let us define $\ell'(f_\theta, G_{y,z}) = \frac{m_{y,z}}{m_{*,z}} \ell(f_\theta, G_{y,z})$ and $\ell'(f_\theta, G_y) = \frac{1}{|\mathbb{Z}|} \sum_{n=1}^{|\mathbb{Z}|} \frac{m_{y,z_n}}{m_{*,z_n}} \ell(f_\theta, G_{y,z_n})$. We then define the cost function for DP as $\mathcal{L}_{DP} = \frac{1}{|\mathbb{Y}||\mathbb{Z}|} \sum_{y \in \mathbb{Y}, z \in \mathbb{Z}} |\tilde{\ell}'(f_\theta, G_{y,z}) - \tilde{\ell}'(f_\theta, G_y)|$. The entire optimization problem is:

$$\begin{aligned} \min_{\mathbf{w}_l} \frac{1}{|\mathbb{Y}||\mathbb{Z}|} \sum_{y \in \mathbb{Y}, z \in \mathbb{Z}} |\tilde{\ell}'(f_\theta, G_{y,z}) - \tilde{\ell}'(f_\theta, G_y)| \\ + \lambda \frac{1}{|\mathbb{Y}_c||\mathbb{Z}|} \sum_{y \in \mathbb{Y}_c, z \in \mathbb{Z}} \tilde{\ell}(f_\theta, G_{y,z}), \\ \text{where } \; \tilde{\ell}(f_\theta, G_{y,z}) = \ell(f_\theta^{l-1}, G_{y,z}) \\ -\eta \nabla_\theta \ell(f_\theta^{l-1}, G_{y,z})^\top \left(\frac{1}{|T_l|} \sum_{d_i \in T_l} \mathbf{w}_l^i \nabla_\theta \ell(f_\theta^{l-1}, d_i) \right). \end{aligned} \tag{4}$$

To find the optimal sample weights for the current task data considering both model accuracy and fairness, we first transform the defined optimization problems of Eqs. 2, 3, and 4 into the form of linear programming (LP) problems.

Theorem 3.4. *The fairness-aware optimization problems (Eqs. 2, 3, and 4) can be formulated as linear programming (LP) problems.*

The loss of each group can be approximated as a linear function, as described in Lemma 3.1. This result implies that the optimization problems, consisting of the loss of each group, can likewise be formulated as LP problems. The proof is in Sec. A.6 of our full paper [45]. We solve the LP problems using linear optimization solvers (e.g., CPLEX [15]). As we add the average loss of the current task data in Eqs. 2, 3, and 4 as a regularization term, the optimal sample weights do not indicate a severely shifted distribution.

3.3 Algorithm

We describe the overall process of fair class-incremental learning in Alg. 1. For the recently arrived current task data, we first perform our fairness-aware sample weighting (FSW) to assign training

Algorithm 1 Fair Class-Incremental Learning

Input: Current task data T_l, previous buffer data $\mathcal{M} = \{\mathcal{M}_1, \ldots, \mathcal{M}_{l-1}\}$, previous model f_θ^{l-1}, loss function ℓ, learning rate η, hyperparameters $\{\alpha, \lambda, \tau\}$, and fairness measure F.

1: **for** each epoch **do**
2: $\quad \mathbf{w}_l^* \leftarrow \textbf{\textit{FSW}}(T_l, \mathcal{M}, f_\theta^{l-1}, \ell, \alpha, \lambda, F)$
3: $\quad g_{\text{curr}} \leftarrow \frac{1}{|T_l|} \sum_{d_i \in T_l} \mathbf{w}_l^{*i} \nabla_\theta \ell(f_\theta^{l-1}, d_i)$
4: $\quad g_{\text{prev}} \leftarrow \nabla_\theta \ell(f_\theta^{l-1}, \mathcal{M})$
5: $\quad \theta \leftarrow \theta - \eta\,(g_{\text{curr}} + \tau\, g_{\text{prev}})$
6: $\mathcal{M}_l \leftarrow$ *Buffer Sample Selection*(T_l)
7: $\mathcal{M} \leftarrow \mathcal{M} \cup \mathcal{M}_l$

Algorithm 2 Fairness-aware Sample Weighting (FSW)

Input: Current task data $T_l = \{\ldots, d_i, \ldots\}$, previous buffer data, $\mathcal{M} = \bigcup_{y \in \mathbb{Y}-\mathbb{Y}_c, z \in \mathbb{Z}} G_{y,z}$, previous model f_θ^{l-1}, loss ℓ, hyperparameters $\{\alpha, \lambda\}$, and fairness measure F.

Output: Optimal training weights $\mathbf{w}_l^*$ for current task data

1: $\ell_G \leftarrow [\ell(f_\theta^{l-1}, G_{1,1}), \ldots, \ell(f_\theta^{l-1}, G_{|\mathbb{Y}|,|\mathbb{Z}|})]$
2: $g_G \leftarrow [\nabla_\theta \ell(f_\theta^{l-1}, G_{1,1}), \ldots, \nabla_\theta \ell(f_\theta^{l-1}, G_{|\mathbb{Y}|,|\mathbb{Z}|})]$
3: $g_d \leftarrow [\ldots, \nabla_\theta \ell(f_\theta^{l-1}, d_i), \ldots]$
4: **switch** F **do**
5: $\quad$ **case** EER: $\mathbf{w}_l^* \leftarrow$ Solve Eq. 2
6: $\quad$ **case** EO: $\mathbf{w}_l^* \leftarrow$ Solve Eq. 3
7: $\quad$ **case** DP: $\mathbf{w}_l^* \leftarrow$ Solve Eq. 4
8: **return** $\mathbf{w}_l^*$

weights that can help learn new knowledge of the current task while retaining accurate and fair memories of previous tasks (Step 2). Next, we train the model using the current task data with its corresponding weights and stored buffer data of previous tasks (Steps 3–5), where η is a learning rate, and τ is a hyperparameter to balance between them during training. The sample weights are computed once at the beginning of each epoch, and they are applied to all batches for computational efficiency [26, 27]. This procedure is repeated until the model converges (Steps 1–5). Before moving on to the next task, we employ buffer sample selection to store a small data subset for the current task (Steps 6–7). Buffer sample selection can also be done with consideration for fairness, but our experimental observations indicate that selecting representative and diverse samples for the buffer, as previous studies have shown, results in better accuracy and fairness. We thus employ a simple random sampling technique for the buffer sample selection in our framework (see Sec. B.15 of our full paper [45] for more details).

Alg. 2 shows the fairness-aware sample weighting (FSW) algorithm for the current task data. We first divide both the previous buffer data and the current task data into groups based on each class and sensitive attribute. Next, we compute the average loss and gradient vectors for each group (Steps 1–2), and individual gradient vectors for the current task data (Step 3). To compute gradient vectors, we use the last layer approximation, which only considers the gradients of the model's last layer, that is efficient and known to be reasonable [5, 24, 26, 27, 29, 42] (see Sec. B.16 of our full paper [45] for details). We then solve linear programming to find the optimal sample weights for a user-defined target fairness measure such as EER (Step 5), EO (Step 6), and DP (Step 7). Since the gradient norm of the current task data is significantly larger than that of the buffer data, we utilize normalized gradients to update the loss of each group and replace the learning rate η with a hyperparameter α in the equations. Finally, we return the current task's sample weights for training (Step 8).

Training with FSW theoretically guarantees model convergence under the assumptions that the training loss is Lipschitz continuous and strongly convex, and that a proper learning rate is used [8, 26, 35]. The computational complexity of FSW is quadratic to the number of current task samples, as CPLEX, which uses simplex-based algorithms, typically exhibits quadratic complexity with respect to the number of variables when solving LP problems in practice [7]. Our empirical results show that for about twelve thousand current task samples, the time to solve an LP problem is a few seconds, which leads to a few minutes of overall runtime for MNIST datasets (see Sec. B.3 of our full paper [45] for details). Since we focus on continual offline training of large batches or separate tasks, rather than online learning, the overhead is manageable enough to deploy updated models in real-world applications. If the task size becomes too large, clustering similar samples and assigning weights to the clusters, rather than samples, could be a solution to reduce the computational overhead.

4 Experiments

In this section, we construct experiments on FSW and address the following research questions: **RQ1** How well can FSW mitigate the unfair forgetting that occurs in class-incremental learning with better accuracy-fairness tradeoff? **RQ2** How does FSW weight the samples? **RQ3** Can FSW be further integrated with fair post-processing techniques?

4.1 Experiment Settings

4.1.1 Metrics. We follow the prior fair continual learning literature to evaluate accuracy and fairness [13, 52].

Average Accuracy. We denote $A_l = \frac{1}{l}\sum_{t=1}^{l} a_{l,t}$ as the accuracy at the lth task, where $a_{l,t}$ is the accuracy of the tth task after learning the lth task. We average A_l over all tasks to obtain the final average accuracy, $\overline{A_l} = \frac{1}{L}\sum_{l=1}^{L} A_l$ where L is the total number of tasks.

Average Fairness. We measure fairness for each task and then average these values to obtain the overall fairness score. Depending on the evaluation scenario, we select one of three disparity measures. In all cases, lower disparity indicates better fairness.
(1) EER computes the average difference in test error rates among classes: $\frac{1}{|\mathbb{Y}|}\sum_{y \in \mathbb{Y}} |\Pr(\hat{y} \neq y | y = y) - \Pr(\hat{y} \neq y)|$. EER targets fairness in predictive quality, aiming for equitable accuracy or error across different prediction outcomes. EER is used as a target metric where a sensitive attribute (z) is not explicitly provided. This scenario includes commonly used class-incremental learning benchmarks where only the class label (y) is available. In such cases, EER serves as a practical proxy for fairness under task imbalance, focusing on equal error across classes.

On the other hand, when z is available, fairness can also be evaluated using group-based metrics like EO and DP, which use both y and z to cover broader social norms.

Table 1: Experimental settings for the five datasets.

Dataset	Size	#Features	#Classes	#Tasks
MNIST	60K	28×28	10	5
FMNIST	60K	28×28	10	5
Biased MNIST	60K	3×28×28	10	5
DRUG	1.3K	12	6	3
BiasBios	253K	128×768	25	5

(2) EO disparity computes the average difference in accuracy among sensitive groups for all classes: $\frac{1}{|\mathbb{Y}||\mathbb{Z}|}\sum_{y\in\mathbb{Y},z\in\mathbb{Z}}|\Pr(\hat{y}=y|y=y,z=z)-\Pr(\hat{y}=y|y=y)|$. EO is preferred when ensuring equal model performance across groups or when true/false positives are equally important.
(3) DP disparity computes the average difference in class prediction ratios among sensitive groups for all classes: $\frac{1}{|\mathbb{Y}||\mathbb{Z}|}\sum_{y\in\mathbb{Y},z\in\mathbb{Z}}|\Pr(\hat{y}=y|z=z)-\Pr(\hat{y}=y)|$. DP is preferred when mitigating data bias or achieving 'blind fairness' by equalizing output distributions.

4.1.2 Datasets. We use a total of five datasets as shown in Table 1. We first utilize commonly used benchmarks for continual image classification tasks, which include MNIST and Fashion-MNIST (FMNIST). Here we regard the class as the sensitive attribute and evaluate fairness with EER disparity. We also use multi-class fairness benchmark datasets that have sensitive attributes [14, 16, 46, 56]: Biased MNIST, Drug Consumption (DRUG), and BiasBios. We consider background color as the sensitive attribute for Biased MNIST, and gender for DRUG and BiasBios. We then use EO and DP disparity to evaluate fairness. We also consider using other datasets in the fairness field, but they are unsuitable for class-incremental learning experiments because either there are only two classes, or it is difficult to apply group fairness metrics (see Sec. B.4 of our full paper [45] for details).

4.1.3 Models and Hyperparameters. Following the experimental setups of [11, 41], we use a two-layer MLP with each 256 neurons for the MNIST, FMNIST, Biased MNIST, and DRUG datasets. For the BiasBios dataset, we use a pre-trained BERT language model [17]. We employ single-head evaluation where a final layer of the model is shared for all tasks [9, 19]. To solve the fairness-aware optimization problems and find optimal sample weights, we use CPLEX, a high-performance optimization solver developed by IBM that specializes in solving linear programming (LP). See Sec. B.17 for experiments comparing CPLEX with another LP solver, HiGHS [23]. For our buffer storage, we store 32 samples per sensitive group for all experiments. For the hyperparameters α, λ, and τ used in our algorithms, we perform cross-validation with a sequential grid search to find their optimal parameters. Model details and hyperparameter set are provided in Sec. B.5 of our full paper [45].

4.1.4 Baselines. Following prior work [3, 58], all baselines are continual learning methods, as other approaches do not fit this scenario. We compare FSW with algorithms from four categories:

Naïve methods. Joint Training assumes access to all the data from previous classes for training, thereby serving as an upper bound in terms of performance; *Fine Tuning* trains a model only on data from new classes, without access to previous data, and thus serves as a lower bound.

State-of-the-art methods. iCaRL [47] performs herding-based buffer selection and representation learning using knowledge distillation loss; *WA* [59] is a model rectification method designed to correct the bias in the model's final fully-connected layer. *WA* uses weight aligning to equalize the norms of the weight vectors over classes; *CLAD* [55] is a representation learning method that disentangles the representation interference between old and new classes.

Sample selection methods. GSS [3] selects a buffer with diverse gradients of samples; *OCS* [58] uses gradient-based similarity, diversity, and affinity scores to rank and select samples for both current and buffer data in a unified manner.

Fairness-aware methods. FaIRL [13] performs fair representation learning by controlling the rate-distortion function.

Our choice of baselines was driven by practical considerations of computational cost and spatial complexity. For instance, while *L2P* [54] achieves strong performance, it leverages a pretrained vision transformer that already embeds extensive prior information, making it less directly comparable to FSW. Similarly, *DER* [57] trains a separate feature extractor for each task, which incurs additional storage overhead and complexity. In contrast, the baselines like *iCaRL* and *WA* are widely recognized as strong and representative baselines in many studies [22, 60, 61], offering a fair and practical comparison. *FairCL* [52] addresses fairness in semantic segmentation tasks arising from the imbalanced class distribution of pixels, but we consider this problem to be unrelated from ours to add the method as a baseline.

4.2 Accuracy and Fairness Results

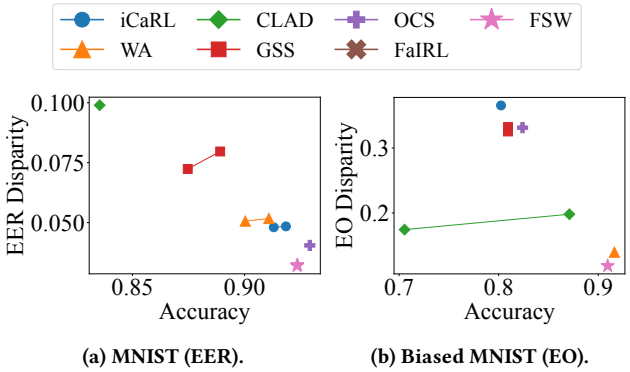
To answer **RQ1**, we compare FSW with other baselines in terms of accuracy and corresponding fairness metrics as shown in Table 2 and Sec. B.6 of our full paper [45]. The accuracy-fairness Pareto front is shown in Fig. 2 and Sec. B.7 of our full paper [45]. Due to the excessive time required to run *OCS* on BiasBios, we are not able to measure the results. Detailed sequential performance results and additional analyses on varying buffer sizes are provided in Sec. B.8 and B.9 of our full paper [45], respectively.

Overall, FSW achieves better accuracy-fairness tradeoff results compared to the baselines across all the datasets. Although FSW does not achieve the best performance in accuracy, FSW shows the best fairness results among the baselines with similar accuracies (e.g., *iCaRL*, *WA*, *CLAD*, *GSS*, and *OCS*) and thus has the best accuracy-fairness tradeoff. We observe that FSW also improves model accuracy while enhancing the performance of underperforming groups for fairness.

The state-of-the-art method, *iCaRL*, generally achieves high accuracy with low EER disparity results. However, *iCaRL*'s nearest-mean-of-exemplars approach for its classification model, the predictions are significantly affected by sensitive attribute values, resulting in high disparities for EO and DP. Although *WA* also performs well, the method adjusts the model weights for the current task classes as a whole, which leads to an unfair forgetting of sensitive groups and unstable results. The closest work to FSW is *CLAD*, which disentangles the representations of new classes and a fixed

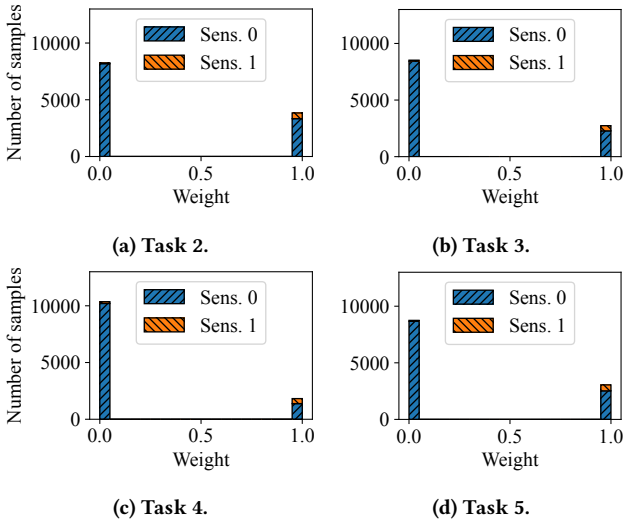
Table 2: Accuracy and fairness results with respect to (1) EER disparity, where class is considered the sensitive attribute for MNIST and FMNIST datasets, and (2) EO disparity, where background color and gender are the sensitive attributes for the Biased MNIST and DRUG datasets, respectively (see DP disparity and the BiasBios dataset results in Sec. B.6 of our full paper [45]). We mark the best and second-best results with bold and underline, respectively, excluding the naïve methods.

Methods	MNIST		FMNIST		Biased MNIST		DRUG		BiasBios	
	Acc.	EER Disp.	Acc.	EER Disp.	Acc.	EO Disp.	Acc.	EO Disp.	Acc.	EO Disp.
Joint Training	.989±.000	.003±.000	.921±.002	.024±.002	.944±.002	.108±.003	.442±.015	.179±.052	.823±.002	.076±.001
Fine Tuning	.455±.000	.326±.000	.451±.000	.325±.000	.449±.001	.016±.002	.357±.009	.125±.034	.420±.001	.028±.002
iCaRL	.918±.005	.048±.003	**.852±.002**	.047±.001	.802±.008	.365±.021	**.444±.025**	.190±.017	**.829±.002**	.084±.003
WA	.911±.007	.052±.006	.809±.005	.088±.003	**.916±.002**	.140±.004	.408±.022	.134±.029	.796±.003	.076±.001
CLAD	.835±.016	.099±.016	.782±.018	.118±.022	.871±.012	.198±.022	.410±.026	.114±.043	.799±.003	.074±.002
GSS	.889±.010	.080±.009	.732±.021	.149±.019	.809±.005	.325±.017	.426±.010	.167±.038	.808±.003	.081±.002
OCS	**.929±.002**	.040±.003	.799±.008	.109±.007	.824±.007	.331±.013	.406±.024	.142±.003	–	–
FaIRL	.558±.060	.273±.018	.531±.032	.289±.019	.411±.012	**.118±.011**	.354±.011	**.060±.021**	.400±.060	**.055±.020**
FSW	.925±.004	**.032±.005**	.824±.006	**.039±.006**	.909±.004	.119±.007	.406±.014	.077±.010	.808±.002	.072±.001



(a) MNIST (EER). (b) Biased MNIST (EO).

Figure 2: Tradeoff results between accuracy and fairness on the MNIST and Biased MNIST datasets. FSW positioned in the lower right corner of the graph, indicating better accuracy-fairness tradeoff results compared to other baselines.



(a) Task 2. (b) Task 3.

(c) Task 4. (d) Task 5.

Figure 3: Distribution of sample weights for EO in sequential tasks of the Biased MNIST dataset.

proportion of conflicting old classes to mitigate imbalanced forgetting across classes. However, the proportion of conflicts may vary by task in practice, limiting *CLAD*'s ability to achieve group fairness. While the two sample selection methods *GSS* and *OCS* store diverse and representative samples in the buffer, these methods sometimes result in an imbalance in the number of buffer samples across sensitive groups. The fairness-aware method *FaIRL* leverages an adversarial debiasing framework combined with a rate-distortion function. While this method theoretically promises improved fairness, it suffers from significant accuracy degradation due to the instability of jointly training the feature encoder and discriminator.

In comparison, FSW explicitly utilizes approximated loss and fairness measures to adjust the training weights for the current task samples, which leads to better model accuracy and fairness. As shown in Fig. 2 and detailed in Sec. B.7 of our full paper [45], FSW consistently dominates the Pareto frontier, occupying the optimal bottom-right region across all evaluated tasks. These results demonstrate that FSW consistently outperforms other baselines and achieves better accuracy-fairness tradeoff.

4.3 Sample Weighting Analysis

To answer **RQ2**, we analyze how our FSW algorithm weights the current task samples at each task in Fig. 3 and Sec. B.10 of our full paper [45]. As the acquired sample weights may change with epochs during training, we show the average weight distribution of sensitive groups over all epochs. Since FSW is not applied to the first task, where the model is trained with only the current task data, we present results starting from the second task.

The result shows that FSW assigns higher weights on average to the underperforming group (Sensitive group 1 in Fig. 3) compared to the overperforming group (Sensitive group 0 in Fig. 3). The weights

Table 3: Average counts of binary (0 or 1) and non-binary (not 0 or 1) sample weights for each optimization task. Since we take averages of all tasks except the first, the sum of (# Binary) and (# Non-binary) is not necessarily an integer.

Dataset (Metric)	MNIST (EER)	FMNIST (EER)	Biased MNIST (EO)	Biased MNIST (DP)
# Binary	11830.8	11997.4	11831.9	11831.6
# Non-binary	3.0	2.6	1.9	2.1

Dataset (Metric)	Drug (EO)	Drug (DP)	BiasBios (EO)	BiasBios (DP)
# Binary	446.5	446.1	25274.7	25273.2
# Non-binary	0.0	0.4	2.0	3.5

Table 4: Accuracy and fairness results on the MNIST and Biased MNIST datasets with or without FSW.

Methods	MNIST		Biased MNIST	
	Acc.	EER Disp.	Acc.	EO Disp.
W/o FSW	$.912_{\pm.004}$	$.051_{\pm.005}$	$\mathbf{.910_{\pm.003}}$	$.126_{\pm.005}$
FSW	$\mathbf{.925_{\pm.004}}$	$\mathbf{.032_{\pm.005}}$	$.909_{\pm.004}$	$\mathbf{.119_{\pm.007}}$

Table 5: Accuracy and more fairness metrics (including EO, DP, and EER Disparity) results on the Biased MNIST dataset without FSW and FSW with respect to EO and DP disparity.

Methods	Acc.	EO Disp.	DP Disp.	EER Disp.
W/o FSW	$\mathbf{.910_{\pm.003}}$	$.126_{\pm.005}$	$.009_{\pm.001}$	$.032_{\pm.002}$
FSW (w.r.t. EO Disp.)	$.909_{\pm.004}$	$\mathbf{.119_{\pm.007}}$	$\mathbf{.008_{\pm.001}}$	$.032_{\pm.003}$
FSW (w.r.t. DP Disp.)	$.904_{\pm.004}$	$.124_{\pm.008}$	$\mathbf{.008_{\pm.001}}$	$\mathbf{.029_{\pm.004}}$

are computed by considering complex forgetting relationships between sensitive groups, which differs from simply assigning higher weights to underperforming groups. Note that the acquired sample weights are mostly close to 0 or 1, which are extreme values. As optimal solutions of a linear program are obtained at the vertices of the feasible region, a convex polytope defined by $\mathbf{w}_I \in [0,1]^{|T_I|}$ (as formulated in Sec. 3.2 of our full paper [45]), each w coincides with its extreme values. However, there are also some values that do not lie at these boundaries (0 or 1). The number of binary (0 or 1) and non-binary (not 0 or 1) samples weights are shown in Table 3.

We also emphasize that if we limit the solution of the optimization problem to binary, which is equivalent to sample selection, the problem would transform into a mixed-integer linear programming problem, which is NP-hard and cannot be solved efficiently. We also observe that FSW assigns zero weight to a significant number of samples, reducing the training data used. This weighting approach provides an additional advantage in enabling efficient model training while retaining accuracy and fairness.

Table 6: Accuracy and fairness (DP disparity) results when combining fair post-processing technique (ϵ-fair) with continual learning methods (*iCaRL*, *WA*, *OCS*, and FSW).

Methods	Biased MNIST		DRUG	
	Acc.	DP Disp.	Acc.	DP Disp.
iCaRL	$.802_{\pm.008}$	$.015_{\pm.001}$	$\mathbf{.444_{\pm.025}}$	$.093_{\pm.009}$
WA	$.916_{\pm.002}$	$.009_{\pm.001}$	$.408_{\pm.022}$	$.067_{\pm.013}$
OCS	$.824_{\pm.007}$	$.035_{\pm.003}$	$.393_{\pm.017}$	$.053_{\pm.012}$
FSW	$.904_{\pm.004}$	$.008_{\pm.001}$	$.405_{\pm.013}$	$.043_{\pm.004}$
iCaRL – ϵ-fair	$\underline{.944}_{\pm.008}$	$\underline{.006}_{\pm.002}$	$\underline{.427}_{\pm.018}$	$\underline{.026}_{\pm.004}$
WA – ϵ-fair	$\mathbf{.953_{\pm.003}}$	$\underline{.006}_{\pm.002}$	$.404_{\pm.021}$	$.044_{\pm.020}$
OCS – ϵ-fair	$\mathbf{.952_{\pm.003}}$	$.032_{\pm.004}$	$.384_{\pm.009}$	$.051_{\pm.002}$
FSW – ϵ-fair	$.906_{\pm.006}$	$\mathbf{.005_{\pm.001}}$	$.405_{\pm.013}$	$\mathbf{.021_{\pm.004}}$

4.4 Ablation Study

To show the effectiveness of FSW on accuracy and fairness, we perform an ablation study comparing the performance of using FSW versus using all current task samples for training with equal weights. The results for the MNIST and Biased MNIST datasets are shown in Table 4. The results for DP disparity and other datasets, which are similar, are in Sec. B.11 of our full paper [45]. As a result, applying sample weighting to the current task data is necessary to improve fairness while maintaining comparable accuracy.

We also investigate how optimizing for one fairness metric influences other metrics in Table 5 and Sec. B.12 of our full paper [45]. Optimizing FSW for a particular fairness metric partially mitigates disparities in other metrics for EO and DP, since both metrics utilize explicitly defined sensitive attributes, though its effectiveness is generally the highest for the targeted metric. In contrast, FSW does not effectively reduce EER disparity, where the sensitive attribute is defined based on the class label rather than an explicitly defined sensitive attribute.

4.5 Integrating FSW with Fair Post-processing

To answer **RQ3**, we emphasize the extensibility of FSW by combining it with a state-of-the-art fair post-processing technique for multi-class tasks, ϵ-fair [16]. Since ϵ-fair supports DP, we only report DP results of the Biased MNIST and DRUG datasets in Table 6. More results and explanations are in Sec. B.13 of our full paper [45]. Comparing original continual learning methods with their integration of ϵ-fair reduces fairness disparity, while accuracy sometimes degrades. This result shows that combining the fair post-processing technique can further improve fairness after model training with continual learning methods. In addition, FSW still shows a better accuracy-fairness tradeoff than existing continual learning methods, even when combined with the fair post-processing technique, compared to existing continual learning methods.

5 Conclusion

We proposed FSW, a fairness-aware sample weighting algorithm for class-incremental learning. Unlike conventional class-incremental learning, we demonstrated how training with all the current task

data using equal weights may result in unfair catastrophic forgetting. We theoretically showed that the average gradient vector of the current task data should not oppose the average gradient vector of a sensitive group to avoid unfair forgetting. We then proposed FSW as a solution to adjust the average gradient vector of the current task data, thereby achieving better accuracy-fairness tradeoff results. FSW supports various group fairness measures by converting the optimization problem into a linear program. In our experiments, FSW consistently outperformed baselines in fairness while maintaining comparable accuracy across diverse datasets from various domains. Future work includes generalizing to multiple sensitive attributes, exploring strategies to further reduce computational overhead, and possible strategies of FSW when data drift occurs. Further discussion and analysis of these future directions are provided in Sec. D of our full paper [45].

Limitations. Efforts to improve fairness inevitably result in a decrease in accuracy. Despite this inherent trade-off, FSW effectively minimizes the accuracy drop and achieves the best accuracy-fairness tradeoff among all other evaluated approaches (see Sec. 4.2 and B.7 of our full paper [45] for more details). Additionally, our theoretical analysis relies on a first-order Taylor approximation, which may introduce approximation errors. While higher-order Taylor approximations could theoretically yield more accurate results, they render the optimization problem non-convex and NP-hard. Empirical analysis further demonstrates that the approximation error diminishes rapidly over training epochs, indicating that our method effectively mitigates approximation inaccuracies introduced by the approximation over time (see Sec. A.3 and B.2 of our full paper [45] for more details, respectively).

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References

[1] Wickliffe C. Abraham and Anthony Robins. 2005. Memory retention – the synaptic stability versus plasticity dilemma. *Trends in Neurosciences* 28, 2 (2005), 73–78.
[2] Ibrahim M. Alabdulmohsin, Jessica Schrouff, and Sanmi Koyejo. 2022. A Reduction to Binary Approach for Debiasing Multiclass Datasets. In *NeurIPS*.
[3] Rahaf Aljundi, Min Lin, Baptiste Goujaud, and Yoshua Bengio. 2019. Gradient based sample selection for online continual learning. In *NeurIPS*. 11816–11825.
[4] Hadis Anahideh, Abolfazl Asudeh, and Saravanan Thirumuruganathan. 2022. Fair active learning. *Expert Systems with Applications* 199 (2022), 116981.
[5] Jordan T. Ash, Chicheng Zhang, Akshay Krishnamurthy, John Langford, and Alekh Agarwal. 2020. Deep Batch Active Learning by Diverse, Uncertain Gradient Lower Bounds. In *ICLR*.
[6] Eden Belouadah, Adrian Popescu, and Ioannis Kanellos. 2021. A comprehensive study of class incremental learning algorithms for visual tasks. *Neural Networks* 135 (2021), 38–54.
[7] Robert E. Bixby. 2002. Solving Real-World Linear Programs: A Decade and More of Progress. *Oper. Res.* 50, 1 (2002), 3–15.
[8] Junyi Chai and Xiaoqian Wang. 2022. Fairness with adaptive weights. In *ICML*. PMLR, 2853–2866.
[9] Arslan Chaudhry, Puneet Kumar Dokania, Thalaiyasingam Ajanthan, and Philip H. S. Torr. 2018. Riemannian Walk for Incremental Learning: Understanding Forgetting and Intransigence. In *ECCV*, Vol. 11215. 556–572.
[10] Arslan Chaudhry, Albert Gordo, Puneet K. Dokania, Philip H. S. Torr, and David Lopez-Paz. 2021. Using Hindsight to Anchor Past Knowledge in Continual Learning. In *AAAI*. 6993–7001.
[11] Arslan Chaudhry, Marc'Aurelio Ranzato, Marcus Rohrbach, and Mohamed Elhoseiny. 2019. Efficient Lifelong Learning with A-GEM. In *ICLR*.
[12] Arslan Chaudhry, Marcus Rohrbach, Mohamed Elhoseiny, Thalaiyasingam Ajanthan, Puneet Kumar Dokania, Philip H. S. Torr, and Marc'Aurelio Ranzato. 2019. Continual Learning with Tiny Episodic Memories. *CoRR* abs/1902.10486 (2019).
[13] Somnath Basu Roy Chowdhury and Snigdha Chaturvedi. 2023. Sustaining Fairness via Incremental Learning. In *AAAI*. 6797–6805.
[14] Nikhil Churamani, Ozgur Kara, and Hatice Gunes. 2023. Domain-Incremental Continual Learning for Mitigating Bias in Facial Expression and Action Unit Recognition. *IEEE Trans. Affect. Comput.* 14, 4 (2023), 3191–3206.
[15] IBM ILOG Cplex. 2009. V12. 1: User's Manual for CPLEX. *International Business Machines Corporation* 46, 53 (2009), 157.
[16] Christophe Denis, Romuald Elie, Mohamed Hebiri, and François Hu. 2023. Fairness guarantee in multi-class classification. arXiv:2109.13642
[17] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *NAACL-HLT*. 4171–4186.
[18] Michele Donini, Luca Oneto, Shai Ben-David, John Shawe-Taylor, and Massimiliano Pontil. 2018. Empirical Risk Minimization Under Fairness Constraints. In *NeurIPS*. 2796–2806.
[19] Sebastian Farquhar and Yarin Gal. 2018. Towards Robust Evaluations of Continual Learning. *CoRR* abs/1805.09733 (2018).
[20] Michael Feldman, Sorelle A. Friedler, John Moeller, Carlos Scheidegger, and Suresh Venkatasubramanian. 2015. Certifying and Removing Disparate Impact. In *KDD*. 259–268.
[21] Moritz Hardt, Eric Price, and Nati Srebro. 2016. Equality of Opportunity in Supervised Learning. In *NIPS*. 3315–3323.
[22] Jiangpeng He. 2024. Gradient reweighting: Towards imbalanced class-incremental learning. In *CVPR*. 16668–16677.
[23] Qi Huangfu and JA Julian Hall. 2018. Parallelizing the dual revised simplex method. *Mathematical Programming Computation* 10, 1 (2018), 119–142.
[24] Angelos Katharopoulos and François Fleuret. 2018. Not All Samples Are Created Equal: Deep Learning with Importance Sampling. In *ICML*, Vol. 80. 2530–2539.
[25] Davinder Kaur, Suleyman Uslu, Kaley J. Rittichier, and Arjan Durresi. 2023. Trustworthy Artificial Intelligence: A Review. *ACM Comput. Surv.* 55, 2 (2023), 39:1–39:38.
[26] KrishnaTeja Killamsetty, Durga Sivasubramanian, Ganesh Ramakrishnan, Abir De, and Rishabh K. Iyer. 2021. GRAD-MATCH: Gradient Matching based Data Subset Selection for Efficient Deep Model Training. In *ICML*, Vol. 139. 5464–5474.
[27] KrishnaTeja Killamsetty, Durga Sivasubramanian, Ganesh Ramakrishnan, and Rishabh K. Iyer. 2021. GLISTER: Generalization based Data Subset Selection for Efficient and Robust Learning. In *AAAI*. 8110–8118.
[28] Dongwan Kim and Bohyung Han. 2023. On the Stability-Plasticity Dilemma of Class-Incremental Learning. In *CVPR*. 20196–20204.
[29] Minsu Kim, Seonghyeon Hwang, and Steven Euijong Whang. 2024. Quilt: Robust Data Segment Selection against Concept Drifts. In *AAAI*. 21249–21257.
[30] James Kirkpatrick, Razvan Pascanu, Neil C. Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A. Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, Demis Hassabis, Claudia Clopath, Dharshan Kumaran, and Raia Hadsell. 2016. Overcoming catastrophic forgetting in neural networks. *CoRR* abs/1612.00796 (2016).
[31] Matthias De Lange, Rahaf Aljundi, Marc Masana, Sarah Parisot, Xu Jia, Ales Leonardis, Gregory G. Slabaugh, and Tinne Tuytelaars. 2022. A Continual Learning Survey: Defying Forgetting in Classification Tasks. *IEEE Trans. Pattern Anal. Mach. Intell.* 44, 7 (2022), 3366–3385.
[32] Jaehoon Lee, Lechao Xiao, Samuel S. Schoenholz, Yasaman Bahri, Roman Novak, Jascha Sohl-Dickstein, and Jeffrey Pennington. 2019. Wide Neural Networks of Any Depth Evolve as Linear Models Under Gradient Descent. In *NeurIPS*. 8570–8581.
[33] Haochen Liu, Yiqi Wang, Wenqi Fan, Xiaorui Liu, Yaxin Li, Shaili Jain, Anil K. Jain, and Jiliang Tang. 2021. Trustworthy AI: A Computational Perspective. *CoRR* abs/2107.06641 (2021).
[34] David Lopez-Paz and Marc'Aurelio Ranzato. 2017. Gradient Episodic Memory for Continual Learning. In *NIPS*. 6467–6476.
[35] Songtao Lu, Ioannis C. Tsaknakis, Mingyi Hong, and Yongxin Chen. 2020. Hybrid Block Successive Approximation for One-Sided Non-Convex Min-Max Problems: Algorithms and Applications. *IEEE Trans. Signal Process.* 68 (2020), 3676–3691.
[36] Zheda Mai, Ruiwen Li, Jihwan Jeong, David Quispe, Hyunwoo Kim, and Scott Sanner. 2022. Online continual learning in image classification: An empirical survey. *Neurocomputing* 469 (2022), 28–51.

[37] Marc Masana, Xialei Liu, Bartlomiej Twardowski, Mikel Menta, Andrew D. Bagdanov, and Joost van de Weijer. 2023. Class-Incremental Learning: Survey and Performance Evaluation on Image Classification. *IEEE Trans. Pattern Anal. Mach. Intell.* 45, 5 (2023), 5513–5533.

[38] Michael McCloskey and Neal J. Cohen. 1989. Catastrophic Interference in Connectionist Networks: The Sequential Learning Problem. Psychology of Learning and Motivation, Vol. 24. 109–165.

[39] Ninareh Mehrabi, Fred Morstatter, Nripsuta Saxena, Kristina Lerman, and Aram Galstyan. 2022. A Survey on Bias and Fairness in Machine Learning. *ACM Comput. Surv.* 54, 6 (2022), 115:1–115:35.

[40] Martial Mermillod, Aurélia Bugaiska, and Patrick BONIN. 2013. The stability-plasticity dilemma: investigating the continuum from catastrophic forgetting to age-limited learning effects. *Frontiers in Psychology* 4 (2013).

[41] Seyed-Iman Mirzadeh, Mehrdad Farajtabar, Razvan Pascanu, and Hassan Ghasemzadeh. 2020. Understanding the Role of Training Regimes in Continual Learning. In *NeurIPS*.

[42] Baharan Mirzasoleiman, Jeff A. Bilmes, and Jure Leskovec. 2020. Coresets for Data-efficient Training of Machine Learning Models. In *ICML*, Vol. 119. 6950–6960.

[43] Jinlong Pang, Jialu Wang, Zhaowei Zhu, Yuanshun Yao, Chen Qian, and Yang Liu. 2024. Fairness Without Harm: An Influence-Guided Active Sampling Approach. In *NeurIPS*.

[44] German Ignacio Parisi, Ronald Kemker, Jose L. Part, Christopher Kanan, and Stefan Wermter. 2019. Continual lifelong learning with neural networks: A review. *Neural Networks* 113 (2019), 54–71.

[45] Jaeyoung Park, Minsu Kim, and Steven Euijong Whang. 2025. Fair Class-Incremental Learning using Sample Weighting. arXiv:2410.01324 [cs.LG]

[46] Preston Putzel and Scott Lee. 2022. Blackbox post-processing for multiclass fairness. *arXiv preprint arXiv:2201.04461* (2022).

[47] Sylvestre-Alvise Rebuffi, Alexander Kolesnikov, Georg Sperl, and Christoph H. Lampert. 2017. iCaRL: Incremental Classifier and Representation Learning. In *CVPR*. 5533–5542.

[48] Yuji Roh, Kangwook Lee, Steven Euijong Whang, and Changho Suh. 2021. FairBatch: Batch Selection for Model Fairness. In *ICLR*.

[49] Yuji Roh, Weili Nie, De-An Huang, Steven Euijong Whang, Arash Vahdat, and Anima Anandkumar. 2023. Dr-Fairness: Dynamic Data Ratio Adjustment for Fair Training on Real and Generated Data. *Transactions on Machine Learning Research* (2023).

[50] Aili Shen, Xudong Han, Trevor Cohn, Timothy Baldwin, and Lea Frermann. 2022. Optimising Equal Opportunity Fairness in Model Training. In *NAACL*. 4073–4084.

[51] Ki Hyun Tae, Hantian Zhang, Jaeyoung Park, Kexin Rong, and Steven Euijong Whang. 2024. Falcon: Fair Active Learning Using Multi-Armed Bandits. *Proceedings of the VLDB Endowment* 17, 5 (2024), 952–965.

[52] Thanh-Dat Truong, Hoang-Quan Nguyen, Bhiksha Raj, and Khoa Luu. 2023. Fairness Continual Learning Approach to Semantic Scene Understanding in Open-World Environments. In *NeurIPS*.

[53] Suresh Venkatasubramanian. 2019. Algorithmic Fairness: Measures, Methods and Representations. In *PODS*. 481.

[54] Zifeng Wang, Zizhao Zhang, Chen-Yu Lee, Han Zhang, Ruoxi Sun, Xiaoqi Ren, Guolong Su, Vincent Perot, Jennifer Dy, and Tomas Pfister. 2022. Learning to prompt for continual learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 139–149.

[55] Shixiong Xu, Gaofeng Meng, Xing Nie, Bolin Ni, Bin Fan, and Shiming Xiang. 2024. Defying Imbalanced Forgetting in Class Incremental Learning. In *AAAI*, Vol. 38. 16211–16219.

[56] Tian Xu, Jennifer White, Sinan Kalkan, and Hatice Gunes. 2020. Investigating Bias and Fairness in Facial Expression Recognition. In *ECCV*, Vol. 12540. 506–523.

[57] Shipeng Yan, Jiangwei Xie, and Xuming He. 2021. Der: Dynamically expandable representation for class incremental learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 3014–3023.

[58] Jaehong Yoon, Divyam Madaan, Eunho Yang, and Sung Ju Hwang. 2022. Online Coreset Selection for Rehearsal-based Continual Learning. In *ICLR*.

[59] Bowen Zhao, Xi Xiao, Guojun Gan, Bin Zhang, and Shu-Tao Xia. 2020. Maintaining Discrimination and Fairness in Class Incremental Learning. In *CVPR*. 13205–13214.

[60] Da-Wei Zhou, Qi-Wei Wang, Zhi-Hong Qi, Han-Jia Ye, De-Chuan Zhan, and Ziwei Liu. 2024. Class-incremental learning: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2024).

[61] Huiping Zhuang, Yue Yan, Run He, and Ziqian Zeng. 2024. Class incremental learning with analytic learning for hyperspectral image classification. *Journal of the Franklin Institute* 361, 18 (2024), 107285.